Chih-Kai Wang

B.S. Student | AI for Mathematics and Verifiable Reasoning

Email | GitHub | Portfolio | Taipei, Taiwan

EDUCATION

National Taipei University of Education | Expected 2028

B.S. student, Mathematics Education Division, Department of Mathematics and Information Education.

RESEARCH FOCUS

AI for Mathematics; automated theorem proving and autoformalization; finite counterexample search; verifiable research agents; evaluation and calibration; reproducible scientific computing; nonlinear differential equations and bifurcation theory.

My work asks what a proof, computation, model output, or review actually licenses us to claim. I build systems that keep the claim, replayable evidence, negative results, and trust boundary inspectable together.

SELECTED OPEN-SOURCE WORK

Finite Witness | JavaScript, WebMCP

- Built a local-first finite graph conjecture laboratory where people and agents configure the same claim, exhaustively search the same bounded graph space, inspect the smallest counterexample, and preserve a deterministic certificate.

RigorGraph | Python

- Built local claim-evidence graphs, deterministic audit, self-contained offline reports, a GitHub Action, and agent skill pack. Public beta; package 1.0.1.

ProofWeave Core v2 | Python, Lean 4

- Built model-independent structured proof certification while keeping formal certificate validity separate from natural-language scope. Core package 2.0.0.

HonestCI | TypeScript

- Built an npm CLI and GitHub Action that detects missing, stale, zero-test, and unexpectedly reduced JUnit evidence. Stable version 1.0.4.

Charlie Alpha 4B | Python, MLX

- Developed an experimental trilingual statistical procedure-selection model for Apple Silicon. The public release preserves its DGP regret improvement and its non-improvements on P-Bench and StatQA.

TECHNICAL SKILLS

Languages: Python, TypeScript | **Formal methods:** Lean 4 / Mathlib (developing)

Research engineering: MLX, LaTeX, JSON Schema, GitHub Actions, cross-platform CI, deterministic validation

Last updated: August 2026